Contents

Introduction

It's clear that, like the introduction of the Internet and mobile phones, artificial intelligence (AI) will have a profound impact on how organizations serve and interact with their customers and users, design and deliver products and services, collaborate with their partners and suppliers, and enable and empower their employees.

What isn't changing is the importance of process. Business processes define how you deliver products and services to your customers; how you work with your suppliers and partners; and how your employees get things done. AI will change the way we build, run, and improve business processes, but the processes themselves won't disappear.

The 2025 State of Process Orchestration and Automation Report, which surveyed IT decision-makers, business decision-makers, and enterprise software architects, shows that 84% of respondents are looking to add more AI capabilities to their business processes over the next three years. If you're planning the same, read this whitepaper to learn about:

- Different types of AI that can be used to automate business processes
- Why process orchestration is critical for the strategic adoption and use of AI
- The emerging role of generative AI agents in business processes
- Three ways to leverage AI tools and services when working with business processes

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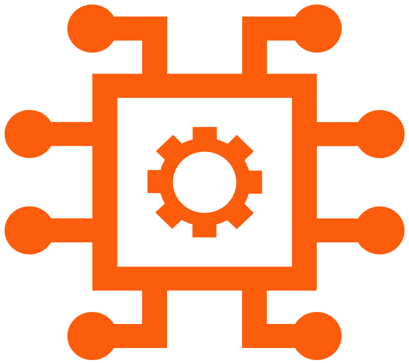
Types of AI used in process automation

From generating content and augmenting decision-making to forecasting future outcomes, different categories of AI bring distinct capabilities—and challenges—to business process automation. Understanding the roles of generative, assistive, and predictive AI is essential to operationalizing these technologies effectively and responsibly.



Generative AI creates content

Generative AI generates new content, data, or solutions based on learned patterns from existing data. Generative AI tools use models such as large language models (LLMs) or generative adversarial networks (GANs) to produce text, images, audio, code, and more. In business contexts, generative AI is increasingly being operationalized to enhance productivity, automate content creation, support decision-making, and drive customer engagement.

Benefits of generative AI:

- Enables rapid creation of content, reducing manual effort
- Accelerates ideation and prototyping across teams and projects
- Enhances personalization at scale in areas such as customer communication and product recommendations

Challenges of generative AI:

- Output quality can vary and may include hallucinated or inaccurate results
- Raises ethical and compliance concerns around copyright, bias, and misinformation
- Requires clear governance to ensure outputs align with brand, policy, and legal standards

Assistive AI augments human work

Assistive AI augments human decision-making by providing contextual guidance, recommendations, or task support in real time. Assistive AI tools offer insights, surface relevant information, and simplify complex workflows to enhance human efficiency and accuracy. Many products now include AI-powered copilots that make smart suggestions based on the context in which the user is working.

Benefits of assistive AI:

- Augments user productivity by offering real-time suggestions and contextual support
- Improves decision-making without removing human oversight or control
- Speeds up complex tasks such as process modeling, troubleshooting, and data entry

Challenges of assistive AI:

- Can become intrusive or disruptive if not well-integrated into user workflows
- Effectiveness depends on the quality of contextual data and user experience design
- May require ongoing tuning to remain relevant and helpful across evolving tasks

Predictive AI optimizes processes

Predictive AI uses machine learning and statistical models to forecast future outcomes based on historical and real-time data. By identifying trends, patterns, and correlations within past events, predictive AI tools anticipate what is likely to happen next. In the context of process automation, predictive AI can analyze process execution data and proactively optimize business processes; for example, by estimating the likelihood of delays or suggesting when human intervention may be required.

Benefits of predictive AI:

- Anticipates outcomes or issues before they occur, enabling proactive action
- Enhances resource planning and performance optimization across operations
- Informs strategic decisions with data-driven foresight and risk assessment

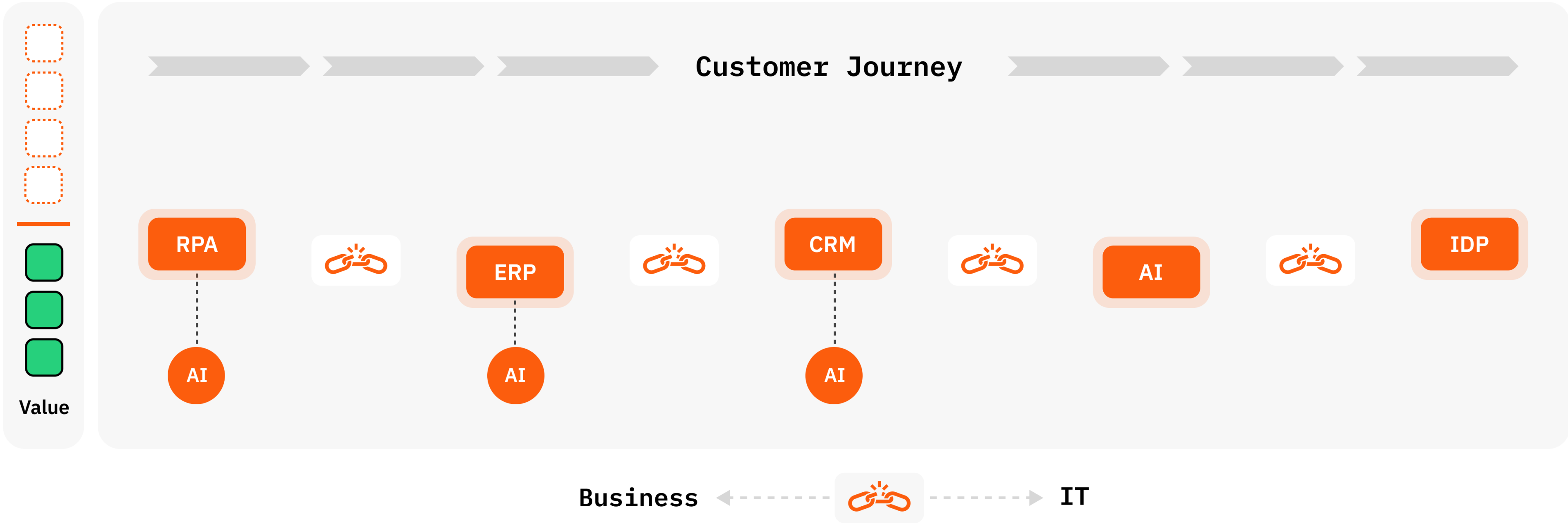
Challenges of predictive AI:

- Requires large, high-quality historical data sets to make accurate predictions
- Predictions can degrade over time due to data drift or changing conditions
- May create overreliance or mistrust if predictions are opaque or inconsistent

Process orchestration is critical to AI adoption

Process orchestration coordinates the steps that make up a business process from start to finish, across a wide variety of people, systems (including AI), and devices. We refer to these as process endpoints. Process orchestration ensures that even the most complex series of steps are executed correctly across all endpoints. In turn, that ensures customers, suppliers, partners, and employees have a smooth experience when they interact with or are affected by business processes.

Many teams that start to experiment with AI tools and services fail to integrate them into business processes alongside the people, systems, and devices that execute process tasks. The result is an automation silo. Automation silos significantly impact the way business gets done. When automated tasks—whether AI-powered or not—aren't integrated into an end-to-end business process, the process becomes inefficient or simply broken. This leads to subpar customer experiences when disconnected, frustrating challenges for employees, operational inefficiencies, and compounding technical debt.



Integrating AI into business processes

There are unlimited opportunities to integrate various forms of AI into business processes to drive efficiency, accuracy, and innovation. Examples include:

- **Image recognition:** Used in insurance claims to automatically assess vehicle damage from uploaded photos or to verify identity through ID document scans
- **Video analysis:** Applied in security or retail to detect anomalies in surveillance footage or analyze customer behavior in stores
- **Computer vision:** Powers quality control in manufacturing by identifying defects in products on assembly lines in real time
- **Knowledge mining:** Extracts actionable insights from large volumes of unstructured documents, such as contracts or legal records, accelerating review cycles
- **Fraud detection:** Flags unusual transaction patterns in banking or claims processing using behavioral analytics and machine learning
- **Speech transcription:** Converts customer service calls or healthcare consultations into text for compliance monitoring and knowledge capture
- **Machine translation:** Enables multilingual customer support or document processing by instantly translating user queries or policy forms
- **Data analysis:** Detects patterns in process execution data to identify bottlenecks, optimize resource allocation, and predict future outcomes

The role of AI agents in process automation

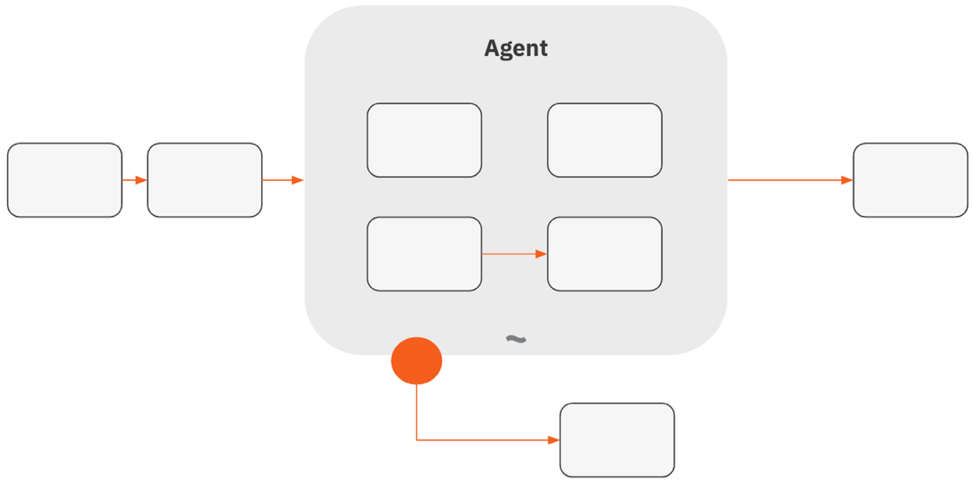
As generative AI tools and services have become widely available, AI agents have emerged as a promising technology for a wide variety of business operations. A generative AI agent is an application that can autonomously and proactively work to achieve a specific goal without human intervention. Agents use AI models (such as LLMs) to plan actions, make decisions, and interact with the environment.

What is agentic process orchestration?

Today, most process orchestration is deterministic, meaning everything that can possibly happen in a process is defined ahead of time, usually by a business process model created in a language such as [Business Process Model and Notation (BPMN)](). Deterministic process orchestration provides a high level of control and ensures processes are predictable and reproducible.

Agents offer the possibility of dynamic process execution, where there isn't a process model or other artifact that defines exactly what will happen each time a process is executed. Instead, agents determine what happens as the process runs, leveraging AI models and runtime data.

Agentic process orchestration blends deterministic and dynamic process execution to provide the best of both worlds. By blending different approaches, agentic orchestration combines the control of a deterministic approach with the flexibility of a dynamic approach. You can model the parts of a process that require high levels of predictability, while leveraging AI agents to handle work that involves creativity and proactive decision-making.



Agentic orchestration BPMN model example with Deterministic Guardrails

How orchestration makes agents more effective

AI agents are powerful, but without orchestration, they lack the coordination, accountability, and reliability needed for real-world, business-critical processes. Agents struggle with:

- **Context switching** and keeping track of multi-step workflows
- **Collaboration** with human decision-makers and other technologies
- **Governance and compliance** that ensure traceability and oversight
- **Scaling** to manage interactions between agents without bottlenecks

To address these challenges, you must integrate AI agents into end-to-end business processes with a solution to orchestrate them. Process orchestration:

- Keeps track of process steps and status and ensures agents have the context and real-time data they need to be most effective
- Enables you to build collaboration points into processes, so agents know when to invoke other technologies and when to flag a human for help
- Provides governance mechanisms such as deterministic process steps, boundaries that agents must work within, and a complete log of what happened in a process
- Can provide fallback process branches and other strategies to mitigate bottlenecks when multiple agents are working together

How agentic process orchestration helps businesses

Agentic process orchestration enables you to automate work that would otherwise be completed by people, while maintaining the flexibility to deliver highly personalized customer experiences. AI agents can replace or augment human tasks, enabling knowledge workers to focus on work that truly requires human attention.

For example, in case management scenarios such as insurance claims, customer complaints, vendor onboarding, loan origination, and legal case investigation, the exact steps that need to be taken are typically not known beforehand. Without AI agents, knowledge workers must handle all aspects of the case, from information gathering to preparing communications. With AI agents, you can increase the level of automation in these scenarios, ensuring humans are only involved when truly needed—for example, when an agent doesn't have the information it needs to make an informed decision or take a particular action.

How to leverage AI in business processes

AI as a siloed capability isn't enough to achieve true business transformation. It must be embedded within business processes to deliver strategic value. When considering how to leverage AI for your business processes, look at three key dimensions:

- How you will orchestrate AI tools and services that are used within processes
- Whether you can accelerate process creation and innovation with assistive AI
- Laying a foundation for AI agents to act more and more autonomously

Orchestrate AI tools and services in processes

According to the [2025 State of Process Orchestration and Automation Report](), 93% of respondents believe AI must be orchestrated across business processes if their organization is to get the maximum benefit from its AI investments. In addition, 90% of respondents said that AI must be orchestrated like any other process endpoint to ensure compliance with regulations.

To start orchestrating AI in business processes, look for a process orchestration and automation solution that:

- [ ] Enables you to design processes that combine AI tools with other process endpoints such as APIs, microservices, enterprise applications, and physical devices
- [ ] Offers pre-built connectors for popular tools and services such as OpenAI, Google Gemini, Hugging Face, and Amazon Bedrock
- [ ] Allows you to customize and extend connectors so you can build in guardrails for their usage within your organization

Put teams on the fast track to automating processes

Assistive AI is transforming the way teams design and automate business processes. AI-powered copilots can put both IT and business teams on the fast track to automating business processes by providing features such as conversational design support, context-aware suggestions, and automated generation of UI elements. Copilots act as real-time collaborators—bridging knowledge gaps, accelerating complex tasks, and making intelligent suggestions tailored to the user's role.

Look for a process orchestration and automation solution with integrated copilots that can:

- [ ] Model business processes without requiring the user to have deep knowledge of languages used for process modeling or data manipulation
- [ ] Make smart recommendations for reusable assets such as connectors and templates
- [ ] Design graphical user interfaces by leveraging a library of UI components

Start building the autonomous enterprise

The autonomous enterprise is a model in which AI agents working in business operations can adapt, make decisions, and execute actions with minimal human intervention. Agents act as digital workers that are embedded in process flows, handling tasks such as document interpretation, decision-making, exception handling, and even dynamic process adaptation.

It isn't possible to achieve this level of autonomy by implementing a collection of stand-alone agents. You need a robust orchestration layer that ensures transparency, compliance, and control. Look for a process orchestration and automation solution that:

- [ ] Can blend deterministic and dynamic orchestration to manage how agents interact, delegate tasks, and hand off to people or other systems as needed
- [ ] Provides end-to-end visibility for business processes as they run and after they are completed, with an audit trail of what happened
- [ ] Has a composable architecture that allows you to swap out different agents and other AI tools without having to redesign processes or rewrite code

Conclusion

AI is poised to become an integral part of how organizations operate—but without structured integration into business processes, its potential will remain untapped. As this whitepaper has shown, realizing the value of AI requires more than isolated experimentation. It demands a strategic orchestration layer that blends the control that deterministic process design provides with the flexibility provided by AI agents.

Organizations that successfully operationalize AI by bringing it into their most business-critical processes will have a competitive edge that will last for years to come, no matter what AI innovations appear next.

Learn more

Camunda is a composable process orchestration and automation solution with an open architecture that's designed to integrate with all types of AI. Learn more about our AI capabilities by trying Camunda for free.

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About Camunda

Camunda enables organizations to orchestrate and automate processes across people, systems, and devices to continuously overcome complexity, increase efficiency, and fully operationalize AI. Built for business and IT, Camunda's leading orchestration and automation platform executes any process at the required speed and scale to remain competitive without compromising security, governance, or innovation. Over 700 companies across all industries, including Atlassian, ING, and Vodafone, trust Camunda with the design, orchestration, automation, and improvement of their business-critical processes to accelerate digital transformation.

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